MECHANICAL PROPERTIES OF FLUIDS

Important formulas and concepts directly from NCERT-

- Liquids and gases can flow and therefore called ‘fluids’.
- Shear stress can change the shape of a solid keeping its volume fixed.
- *Shearing stress* of fluids is about million times smaller than that of solids.

❖ **Pressure** –

- Smaller the area on which the force acts, greater is the impact. the impact is “pressure”.
- ‘Average pressure’ is defined as normal force acting per unit area.

$$P = F/A$$

  Pressure – scalar quantity , SI unit-Nm^{-2} ,dimension-$[ML^{-1}T^{-2}]$
- 1 atm = 1.013×10^{5} Pa
- The liquid is largely incompressible and its density is therefore, **nearly constant at all pressures.**
- **Gases** exhibit a **large variation in densities with pressure**.
- Density of water is 1×10^{3} at **4°C** (277K)
- Relative density = density / density of water at 4°C.
- Relative density of aluminium is 2.7

- **Pascal’s law** –
  - Pressure in a fluid at rest is same at all points if they are at same height.
  - ***Pressure exerted is same in all directions in a fluid at rest .***
  - The force against any area within a fluid at rest and under pressure is normal to the area, regardless of the orientation of area.
- **Variation of pressure with depth-**
  - The pressure P ,at depth ‘h’ below the surface of liquid open to the atmosphere ,

$$P = P_a + \rho gh$$

    The excess of pressure , $(P-P_a)$ at depth h is called gauge pressure at that point.
  - The area of cylinder is not appearing in the expression, thus height of fluid column is important **not base area or shape of container.**

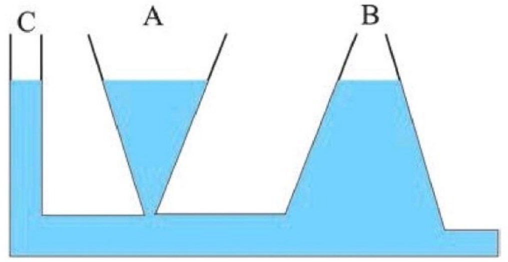


Fig 9.4 *Illustration of hydrostatic paradox. The three vessels A, B and C contain different amounts of liquids, all upto the same height.*

- **Atmospheric pressure and gauge pressure-**
  - The mercury column in the barometer has a height of 76 cm at sea level equivalent to 1atm.

    1 atm = 76 cm of Hg
  - A pressure equivalent of 1 mm is called a torr.

    1mmHg = 1torr = 133 Pa
  - The mm of Hg and torr are used in medicine and physiology. In metereology , a common unit is the bar and millibar.

    1 bar = 10^5 Pa
  - In reality, the density of air decreases with height.
- **Hydraulic brakes** –
  - Whenever external pressure is applied on any part of a fluid contained in a vessel , it is transmitted undiminished and equally in all directions. This is another form of **pascal's law.**

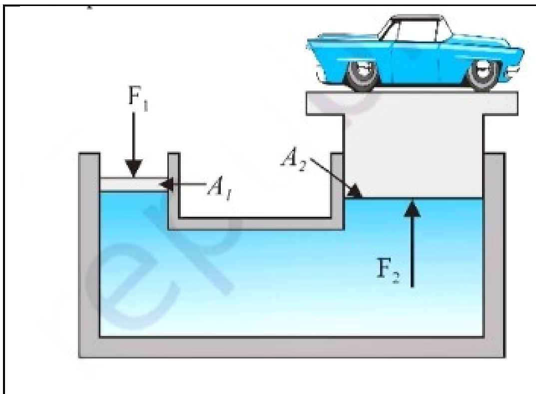


Fig 9.6 *(b) Schematic diagram illustrating the principle behind the hydraulic lift, a device used to lift heavy loads.*

$$F_2 / F_1 = A_2 / A_1$$

The applied force has been increased by a factor of A_2 / A_1 , and this factor is the '**Mechanical advantage'.**

❖ **Streamline flow –**

- The study of fluids in motion is known as 'fluid dynamics' .
- The flow of fluid is said to be **steady** if at any given point,the velocity of each passing fluid particle remains constant in time.
- This doesn't mean that the velocity at different points in space is same.
- The velocity of a particular particle may change as it moves from one point to another.
- The path taken by a fluid particle under a steady flow is a **streamline.**it is defined as the curve whose tangent at any point is in the direction of fluid velocity at that point.

- **Equation of continuity-**

The mass of liquid flowing out equals the mass flowing in,holds in all cases .

For flow of incompressible fluids,

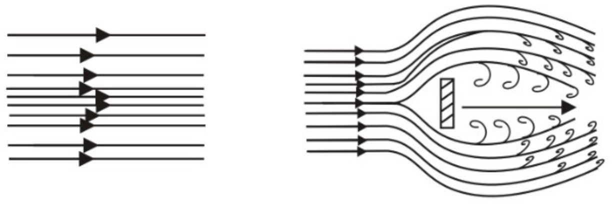
$$A_1v_1 = A_2v_2$$

- It is a statement of **conservation of mass** in flow of incompressible fluids.

$$Av = \text{constant}$$

Av gives the flow rate or volume flux .

- Beyond a limiting value , called **critical speed** ,flow loses steadiness and becomes **turbulent.**



(a) (b)

Fig. 9.8 *(a) Some streamlines for fluid flow. (b) A jet of air striking a flat plate placed perpendicular to it. This is an example of turbulent flow.*

❖ Bernoulli's Principle–

- Based on '**conservation of energy'.**
- Bernoulli's equation,

$$P + \tfrac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

As we move through a streamline , pressure, kinetic energy per unit volume and potential energy per unit volume remains constant.

- It is ideally applies to **non – viscous fluids** .
- Fluids must be **incompressible.**
- Doesn't hold for **non – steady or turbulent flow .**
- When a fluid at rest ;its velocity is zero everywhere.

Eq^n becomes , $(P_1-P_2) = \rho g(h_2-h_1)$

Applications =

[1] Speed of efflux –

[torricelli's law]

$$\mathbf{v = (2gh)^{1/2}} \qquad [\,P = P_a\,]$$

[2] Venturimeter –

- It is a device to measure flow **speed of incompressible fluid**.
- The principle behind this meter has many applications –

1)The carburetor of automobile has a venturi channel (nozzle) through which air flows with a high speed.The pressure is then lowered at the narrow neck and the petrol (gasoline) is sucked up in the chamber to provide correct mixture of air to fuel necessary for combustion.

2)Filter pumps or aspirants , Bunsen burners ,atomisers and sprayers used for perfumes or spray insectisides.

[3] Blood Flow and Heart Attack –

- Bernoulli's principle heps in explaining blood flow in artery.

[4] Dynamic Lift –

- It is the force that acts on a body, by virtue of its motion through a fluid .
- In many games , ball deviates from its trajectory ,This deviation can be partly explained on the basis of Bernoulli's principle.

MAGNUS EFFECT –

The diff. in velocities of air results in pressure difference b/w upper and lower faces and there is net upward force on the ball.the dynamic lift due to spinning is called magnus effect.

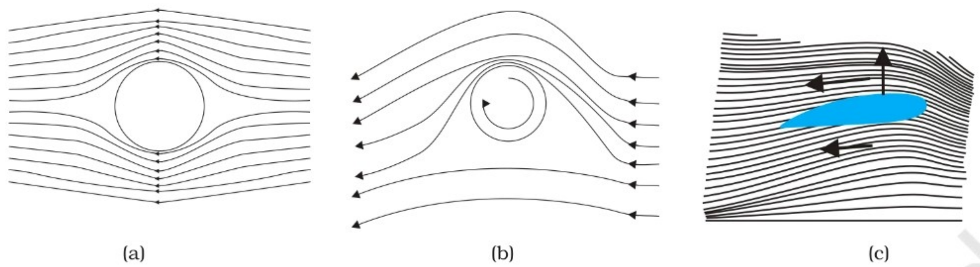


Fig 9.11 *(a) Fluid streaming past a static sphere. (b) Streamlines for a fluid around a sphere spinning clockwise. (c) Air flowing past an aerofoil.*

❖ **Viscosity** –

➢ Most fluids are not ideal and offer some resistance to fluid motion like an internal friction,this is viscosity.

➢ This force exists when there is relative motion b/w layers of liquid .

➢ The coefficient of viscosity for a fluid defined as **ratio** of **shearing stress to strain rate.**

$$\eta = Fl / vA$$

➢ SI unit is poiseiulle (Pl) .

➢ Units – N s m^{-2} or Pa s.

➢ Dimensions – [$ML^{-1}T^{-1}$]

➢ Blood is more thicker (more viscous) than water.
Further,the relative viscosity (η / η_{water}) of blood remains constant b/w 0°C and 37°C.

- Viscosity of **liquids decrease**s with temp,while it **increases** in the case of **gases**.
  - **Stokes' law** –

    Viscous force, **$F = 6\Pi\eta r\upsilon$**
    - Viscous force is directly prop to velocity and opposite to the direction of motion.
    - ***Terminal velocity*** , **$\upsilon_t = 2\ r^2\ (\rho - \sigma)g\ /\ (9\eta)$**

      ρ and σ are mass densities of sphere and fluid respectively.

- **Surface Tension** –
  - As liquids have no definite shape but have a definite volume , *they acquire a free surface* when poured in a container . *these surfaces posses some additional energy.*this phenomenon is surface tension and it is concerned with only liquids as gases do not have free surfaces.
  - A liquid tends to have least surface due to surface tension.
  - **Surface tension = Surface energy / area**

    = force per unit length exerted by fluid
  - For a film , if area is dl then,

    Surface energy = S × 2dl (film has 2 surfaces)
  - Surface tension of liquid air interface ,

    $S = (W/2l) = (mg/2l)$ [for glass plate]
    - **Angle of contact** –
      - Angle b/w tangent to the liquid surface at point of contact and solid surface inside the liquid .

If angle of contact is an **obtuse angle**	If angle of contact is an **acute angle**
• **Water leaf interface** • **Liquid doesn't wet the solid.** • **Happens with water on a waxy or oily surface , and with mercury on any surface.**	• **Water plastic interface** • **Liquid wet the surface.** • **Happens for water on glass or on plastic and for kerosene oil on virtually anything.**

- Soaps , detergents and dying substances are wetting agents,when they are added the angle of contact becomes so small so these may penetrate well and become effective.
- Water proofing agents create a large angle of contact b/w water and fibres.

- **Drops and bubbles** –
  - Rain drop is spherical in shape due to surface tension because spherical shape has less area so less surface energy
  - In general,for a liquid gas interface, convex side has a higher pressure than the concave side.
  - For drop, pressure difference,

(Pi – Po = 2S/r)

For bubble ,

Pi – Po = (4S/r)

- **Capillary rise –**

Height of rise in capillary = 2S/ρgr

(r = radius of curvature of liquid surface)

H = 2Scosɵ/ρgR

(R = radius of capillary tube)

SUMMARY

1. The basic property of a fluid is that it can flow. The fluid does not have any resistance to change of its shape. Thus, the shape of a fluid is governed by the shape of its container. 2. A liquid is incompressible and has a free surface of its own. A gas is compressible and it expands to occupy all the space available to it. If Fis the normal force exerted by a fluid on an area A then the average pressure P 3. is defined as the ratio of the force to area F/A
4. The unit of the pressure is the pascal (Pa). It is the same as N m². Other common units of pressure are 1 atm = 1.01×105 Pa 1 bar = 105 Pa 1 torr 133 Pa = 0.133 kPa 1 mm of Hg= 1 torr 5. Pascal's law states that: Pressure in a fluid at rest is same at all points which are at the same height. A change in pressure applied to an enclosed fluid is transmitted undiminished to every point of the fluid and the walls of the containing vessel. 6. The pressure in a fluid varies with depth h according to the expression P= P + pgh where p is the density of the fluid, assumed uniform. 7. The volume of an incompressible fluid passing any point every second in a pipe of non uniform crossection
8. *Bernoulli's principle* states that as we move along a streamline,the sum of the pressure (P),kinetic energy per unit volume ($\rho v^2/2$) and the potential energy per unit volume (ρgy) remains a constant = constant. P + ρv2 /2 + ρgy = constant The equation is basically the conversion of energy applied to non viscous fluid motion in steady state.There is no fluid which have zero viscosity, so the above statement is true only approximately. The viscosity is like friction and converts the kinetic energy to heat energy. 9. Though shear strain in a fluid does not require shear stress, when a shear stress is applied to a fluid, the motion is generated which causes a shear strain growing with time. The ratio of the shear stress to the time rate of shearing strain is known as coefficient of viscosity, η. where symbols have their usual meaning and are defined in the text. 10. Stokes' law states that the viscous drag force F on a sphere of radius a moving with velocity through a fluid of viscosity is, F 6πηαν. 11. Surface tension is a force per unit length (or surface energy per unit area) acting in the plane of interface between the liquid and the bounding surface. It is the extra energy that the molecules at the interface have as compared to the interior.

POINTS TO PONDER

1. Pressure is a *scalar quantity*. The definition of the pressure as "force per unit area" may give one false impression that pressure is a vector. The "force" in the numerator of the definition is the component of the force normal to the area upon which it is impressed. While describing fluids as a concept, shift from particle and rigid body mechanics is required. We are concerned with properties that vary from point to point in the fluid.
2. One should not think of pressure of a fluid as being exerted only on a solid like the walls of a container or a piece of solid matter immersed in the fluid. Pressure exists at all points in a fluid. An element of a fluid (such as the one shown in Fig. 9.4) is in equilibrium because the pressures exerted on the various faces are equal.
3. The expression for pressure
   $P = P_a + \rho gh$
   holds true if fluid is incompressible. Practically speaking it holds for liquids, which are largely incompressible and hence is a constant with height.
4. The gauge pressure is the difference of the actual pressure and the atmospheric pressure.
   $P - P_a = P_g$
   Many pressure-measuring devices measure the gauge pressure. These include the tyre pressure gauge and the blood pressure gauge (sphygmomanometer).
5. A streamline is a map of fluid flow. In a steady flow two streamlines do not intersect as it means that the fluid particle will have two possible velocities at the point.
6. Bernoulli's principle does not hold in presence of viscous drag on the fluid. The work done by this dissipative viscous force must be taken into account in this case, and P_2 [Fig. 9.9] will be lower than the value given by Eq. (9.12).
7. As the temperature rises the atoms of the liquid become more mobile and the coefficient of viscosity, η falls. In a gas the temperature rise increases the random motion of atoms and η increases.
8. Surface tension arises due to excess potential energy of the molecules on the surface in comparison to their potential energy in the interior. Such a surface energy is present at the interface separating two substances at least one of which is a fluid. It is not the property of a single fluid alone.

Important points from NCERT SUMMARY AND POINTS TO PONDER –

- **Viscosity is like friction and coverts kinetic energy to heat energy .**
- **Surface Tension is the extra energy that the molecules at the interface have as compared to interior.**
- The expression ,

  $$P = Pa + \rho gh$$

  Hold true if fluid is incompressible and hence constant with height.
- Devices measure the gauge pressure
  Tyre pressure gauge and blood pressure gauge (sphygmomanometer)

- **In a steady flow, two styreamlines do not intersect as it means that fluid particle will have two possible velocities at The point.**

- **As Temperature Increases,**

  for liquid , η decreases

  For gases , η Increases